

High energy $\gamma\gamma$ interactions at the LHeC

Laurent Forthomme^a Hamzeh Khanpour^a Krzysztof Piotrzkowski^a Yuji Yamazaki^b

^a*Faculty of Physics and Applied Computer Science, AGH University of Kraków, Al. Mickiewicza 30, 30-055 Kraków, Poland.*

^b*Physics Department, Graduate School of Science, Kobe University, 1-1 Rokkodai-cho, Nada-ku, Kobe, Japan.*

E-mail: laurent.forthomme@cern.ch, hamzeh.khanpour@cern.ch,
krzysztof.piotrzkowski@cern.ch, yamazaki@phys.sci.kobe-u.ac.jp

ABSTRACT: The future Large Hadron-electron Collider (LHeC) is to operate at the center-of-mass energy of 1.2 TeV and is expected to deliver an integrated electron-proton luminosity of about 1 ab^{-1} . Thanks to the high luminosity and a clean experimental environment, the LHeC will open unique research directions. The aim of this paper is to present a comprehensive survey of high energy photon-photon interactions at the LHeC, for the $\gamma\gamma$ center-of-mass energy of up to 1 TeV. The scientific potential of investigating such photon-photon processes is discussed by considering signals and principal backgrounds for various $\gamma\gamma$ processes, including in particular the exclusive production of pairs of W and Z bosons as well as pairs of tau leptons and of charged supersymmetric particles.

Contents

1	Introduction	1
2	Equivalent Photon Approximation and $\gamma\gamma$ luminosity spectra	3
3	Exclusive production of muon (and electron) pairs	4
4	Photon-photon factory of τ pairs	7
5	Exclusive production of W-boson pairs	10
6	Unique Z boson laboratory	12
7	Photoproduction of top quark-antiquark pairs	13
8	Two-photon exclusive production of supersymmetric pairs	15
9	Light-by-light scattering	18
10	Production of the Higgs boson via photon-photon fusion	21
11	Summary and conclusions	24
A	$\gamma\gamma$ luminosity spectrum	25
B	$\gamma\gamma$ cross-sections	25

1 Introduction

The proposed Large Hadron-electron Collider (LHeC) at CERN will open fascinating fields of research [1, 2]. In this article, we discuss the perspectives of its particularly exciting and unique research area of high-energy $\gamma\gamma$ interactions. In case of electron-proton collisions, this new collider will operate at the centre-of-mass energy $\sqrt{s} = 1.2$ TeV for the electron and proton beam energies, $E_e = 50$ GeV and $E_p = 7$ TeV, respectively. As a result, the energies of photon-photon collisions will often exceed the onset energy for electroweak interactions and might even approach the TeV scale. That and a very high nominal LHeC luminosity, in excess of 10^{34} cm⁻²s⁻¹, will allow for original and stringent tests of predictions of the Standard Model (SM) as well as to look for unique manifestations of physics beyond the SM. The expected total integrated luminosity of LHeC is about 1 ab⁻¹ [1, 2]. In addition, a stage one LHeC has recently been proposed to operate during Run5, concurrently with the hadron-hadron experiments at the HL-LHC – the total integrated luminosity at

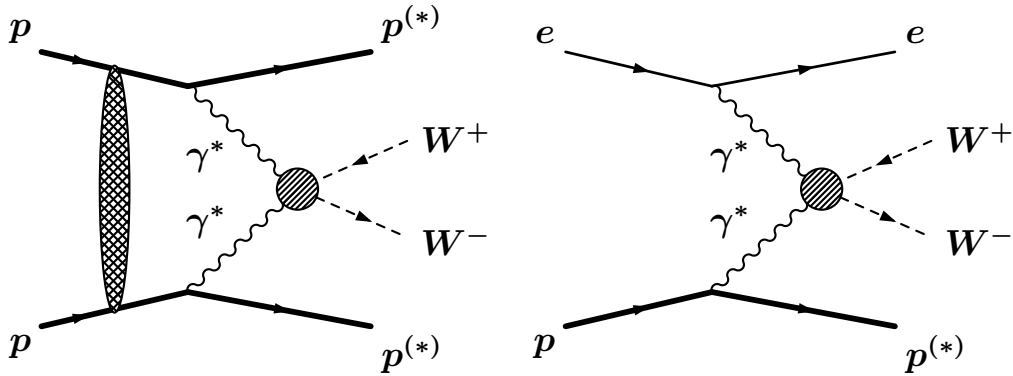


Figure 1. Feynman diagrams representing the (semi-)exclusive production of W -boson pairs via photon-photon fusion at the LHC (left) and future LHeC (right). In the pp diagram the additional exchange between protons (hatched "band") represents the hadronic re-scattering, absent in ep collisions. The asterisks indicate a possibility of the proton dissociation in the semi-exclusive (inelastic) production.

$\sqrt{s} = 0.75$ TeV is expected to be about 250 fb^{-1} [3]. Below, if not explicitly indicated otherwise, the results are given for the $\sqrt{s} = 1.2$ TeV case.

In stark contrast to the working conditions of the pp experiments at the (HL-)LHC, the experimental conditions in the LHeC will be very favourable, especially for the research discussed here. At the LHeC, the event pileup will be negligible and in addition, some sources of overwhelming backgrounds, as due to the Drell-Yan process, for example, will be absent. Moreover, the detection efficiency will be further improved at the LHeC thanks to the use of a "trigger-less" data-streaming, allowing to register also final states which does not have striking features. Finally, as the electron-photon coupling is elementary and there is no suppression of two-photon production due to hadronic re-scattering, therefore, the associated significant theoretical and experimental uncertainties will not intervene (see the Feynman diagrams shown in Fig. 1). It should also be noted that measurements of high-energy two-photon interactions at the LHC were originally proposed to be performed using dedicated, very forward proton detectors to cope with the huge backgrounds in proton-proton collisions [4, 5]. However, pioneering studies $\gamma\gamma$ in the LHC were done without the help of such detectors – it was possible, but only for the exclusive two-photon production of pairs of charged leptons [6]. Only recently were the first measurements reported using very forward proton detectors [7, 8], but they still suffer from the enormous event pile-up at the LHC. As discussed in the following, thanks to its very "clean" environment, the LHeC will allow for wide extension of the scientific scope of high-energy $\gamma\gamma$ research at CERN.

2 Equivalent Photon Approximation and $\gamma\gamma$ luminosity spectra

In the *Equivalent Photon Approximation* (EPA) [9] calculations of cross-sections for the two-photon interactions in electron-proton collisions are factorized (in a similar fashion as in the partonic framework of hadron-hadron collisions) and a cross-section σ_{ep} of a given interaction at the LHeC, proceeding via photon-photon fusion, can be accurately calculated by a convolution of the equivalent photon fluxes for electrons and protons (see App. A), Φ_e and Φ_p , respectively, multiplied by the proper photon-photon cross-section $\sigma_{\gamma\gamma}$:

$$\sigma_{ep} = \int dy_e dy_p \Phi_e(y_e) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma} \sigma_{\gamma\gamma}, \quad (2.1)$$

where the respective photon fractional energies $y_e = E_{\gamma(e)}/E_e$, $y_p = E_{\gamma(p)}/E_p$, W is the $\gamma\gamma$ centre-of-mass energy, and the photon-photon luminosity spectrum $S_{\gamma\gamma}$ is equal to the convolution of fluxes for a given W (for more details see App. A).

In the derivation of Eq. (2.1) it is assumed that $\sigma_{\gamma\gamma}$ is not sensitive to the photon virtualities, so the fluxes could be integrated over the photon Q^2 :

$$\Phi_\gamma(y) = \frac{\alpha}{\pi y} \int \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right],$$

where α is the fine-structure constant and F_E , F_M are the electric and magnetic form-factors, respectively [9]. For the electrons $F_E = F_M = 1$, whereas for the protons, assuming *elastic* scattering and the dipole approximation, $F_M = G_M^2$ and $F_E = (4M_p^2 G_E^2 + Q^2 G_M^2)/(4M_p^2 + Q^2)$, where $G_E^2 = G_M^2/7.78 = (1 + Q^2/0.71 \text{ GeV}^2)^{-4}$ and M_p is the proton mass. In the integral, $Q_{\max}^2$ is set according to the experimental conditions and $Q_{\min}^2 = M^2 y^2/(1-y)$, where M is equal to the electron or proton mass, respectively.

In the case of the semi-exclusive *inelastic* production when the proton does not survive the interaction and dissociates into a state of the invariant mass $M_N > M_p$, $F_E = \int dx F_2/x$ and $F_M = \int dx F_2/x^3$, where $F_2(x, Q^2)$ is the proton structure function and $M_N^2 - M_p^2 = Q^2(1/x - 1)$. In order to calculate $F_E(Q^2)$ and $F_M(Q^2)$, we use the ALLM parameterisation for the proton structure function, which gives a good description of the DIS data for the kinematical region of $10^{-6} < x < 0.85$ and $\text{GeV } 0 \leq Q^2 < 5000 \text{ GeV}^2$ [10, 11]. We also examined other very recent proton PDFs, namely CT18 [12] and NNPDF4.0 [13], for their extended, allowed kinematical region, *i.e.* $Q_{\min}^2 > Q_0^2$, and very consistent results with the ALLM parametrisation are achieved. In such an inelastic production, the minimal photon virtuality is larger than in the elastic case, as $Q_{\min}^2 = (M_N^2/(1-y) - M_p^2)y$, and Q^2 can attain very large values¹.

In Fig. 2 the LHeC elastic and inelastic luminosity spectra $S_{\gamma\gamma}$, and their integrals $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma}$, are shown. Such integrals effectively give a fraction of the ep luminosity available for $\gamma\gamma$ collisions, above a given minimal centre-of-mass energy W_0 . It is remarkable that this fraction for the sum of elastic and inelastic contributions will reach 50% for $W_0 = 10 \text{ GeV}$ at $\sqrt{s} = 1.2 \text{ TeV}$ and about 25% at $\sqrt{s} = 0.75 \text{ TeV}$. This effectively converts the

¹In contrast to the elastic case, where the photon virtuality will very rarely exceed 1 GeV^2 due to a very strong flux suppression at large Q^2 by the elastic form-factors of the proton.

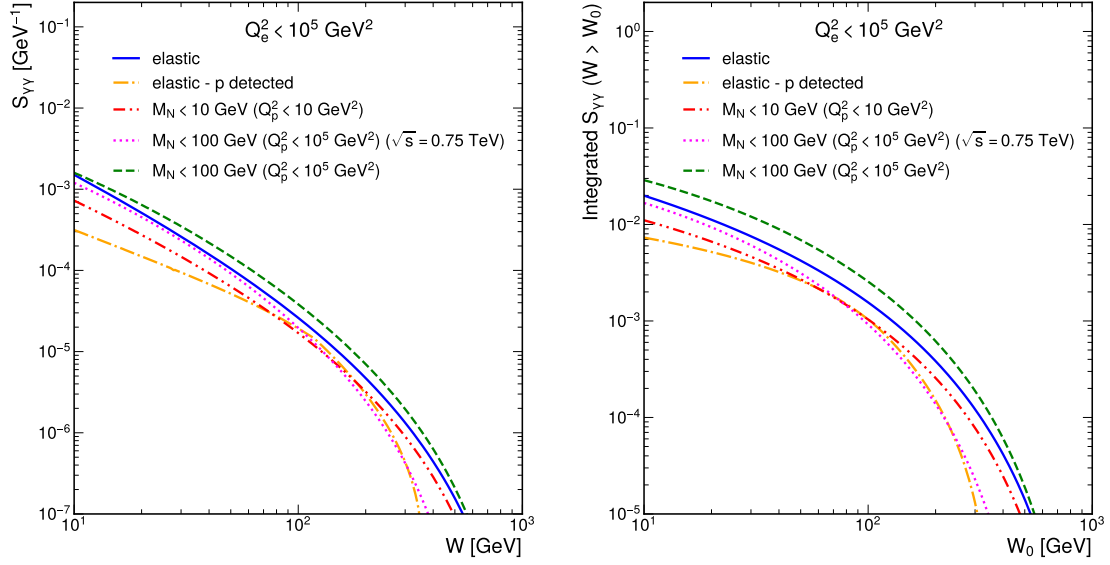


Figure 2. (left) The elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC, assuming $Q_{e,max}^2 = 10^5 \text{ GeV}^2$ and the proper kinematical limits for the minimal photon virtualities (see text above) as well as the tagged elastic spectrum (assuming the proton detection corresponding to $0.01 < y_p < 0.1$) and two inelastic luminosity spectra where the maximum dissociative mass of 10 and 100 GeV is assumed for $Q_{p,max}^2 = 10$ and 10^5 GeV^2 , respectively; (right) the integrated luminosity spectra, $\int dW S_{\gamma\gamma}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 . An inelastic case for $\sqrt{s} = 0.75 \text{ TeV}$ is also shown.

LHeC into a powerful $\gamma\gamma$ collider. One may note that the inelastic contribution dominates over the full W range if very large dissociative masses (and large photon virtualities at the proton vertex) are allowed. On the other hand, the $\gamma\gamma$ luminosity for "quasi-elastic" events (i.e. for $M_N < 10 \text{ GeV}$ and $Q_p^2 < 10 \text{ GeV}^2$) is smaller than for the elastic ones.

In elastic events, a significant fraction of very forward scattered protons can be detected in dedicated near-beam detectors as shown in Fig. 2, allowing to improve reconstruction of event kinematics, which is particularly useful for the final states containing neutrinos. However, because of the lack of event pileup and low backgrounds in general, the detection of these protons is often not necessary to improve the purity of the data samples. Therefore, as discussed in the following, in many cases a large fraction of the total $\gamma\gamma$ luminosity should be successfully used for studying photon-photon interactions at the LHeC.

3 Exclusive production of muon (and electron) pairs

The two-photon production of lepton pairs, as shown in Fig. 3, is very important at the LHeC. The case of tau pairs is presented in the next section; here, we discuss the case of muon pairs, which, together with the e^+e^- pairs, is particularly relevant for having a very good control of the $\gamma\gamma$ measurements at the LHeC.

The cross sections for the elastic and inelastic two-photon production of muon pairs at the LHeC, for two values of the minimal $\gamma\gamma$ centre-of-mass energy W_0 are shown in

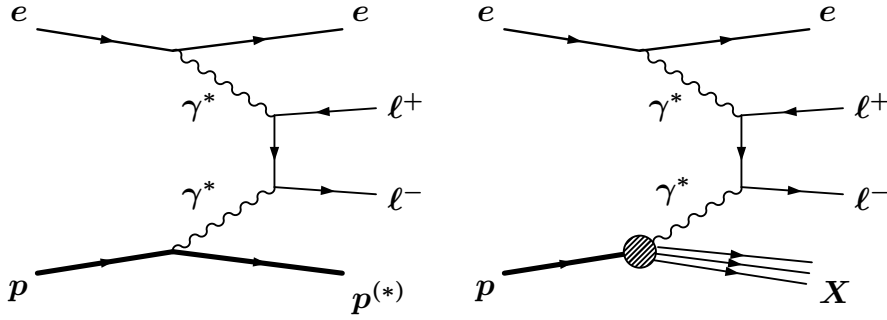


Figure 3. Feynman diagrams for the elastic (left), and for the inelastic (right) two-photon production of lepton pairs at the LHeC.

Table 1. Cross-sections (in pb) for elastic production of $\mu^+\mu^-$ at the LHeC, calculated using the EPA approach for a minimum dimuon invariant mass $W_0 = 10$ and 100 GeV, assuming $Q_{e/p}^2 < 10$ GeV² for the photon virtuality at the electron/proton vertex. The cross-section values calculated using GRAPE and the CEPGEN implementation of LPAIR are also presented for comparison.

σ^{el} [pb]	$Q_{e/p}^2 < 10$ GeV ²	
	$W_0 = 10$ GeV	$W_0 = 100$ GeV
EPA	120.12	0.193
CEPGEN (LPAIR, dipole FF)	118.31	0.191
GRAPE (BH, dipole FF)	121.37	0.193
GRAPE (BH+ISR)	127.61	0.198

Tab. 1 and 2, respectively. The cross sections for $W_0 = 10$ GeV are large, allowing testing with a high precision understanding of two-photon production at the LHeC and a number of various experimental aspects, in particular the separation of the elastic and inelastic contributions using acoplanarity angle, for example. The cross sections for $W_0 = 100$ GeV are significant enough to make such tests also at large $\gamma\gamma$ centre-of-mass energies, which is particularly important with respect to the precision studies of W -boson pairs discussed in Sec. 5.

The cross sections in Tab. 1 and 2 were calculated using the EPA approach, as well as two Monte Carlo event generators using matrix elements (ME), GRAPE [14] and LPAIR [15] (as implemented in CEPGEN [16]). Excellent agreement is observed for elastic production, at a per-cent level. It should be noted that this was possible only once the proper formula for W was applied in the EPA (see Eq. (A.5)), due to the large Q_e^2 involved. Moreover, the inclusion in GRAPE of the effects due to the initial state radiation (ISR) also has a very small impact. Finally, adding to the two-photon amplitudes (named the Bethe-Heitler, BH, contribution) those including Z -exchanges has negligible impact. The differences in inelastic cross-sections in Tab. 2 are larger, also due to the dependence on the assumed proton structure functions. In conclusion, these comparisons show that the EPA can be

Table 2. Cross-sections (in pb) for inelastic production of $\mu^+\mu^-$ at the LHeC, calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 = 10$ and 100 GeV, assuming $Q_{e,p}^2 < 10$ GeV² for the electron/proton vertex, and for $M_N < 3$ GeV. The cross-section values calculated using GRAPE and the CEPGEN implementation of LPAIR are also presented for comparison.

σ^{inel} [pb]	Structure func. modelling	$Q_{e,p}^2 < 10$ GeV ² , $M_N < 3$ GeV	
		$W_0 = 10$ GeV	$W_0 = 100$ GeV
EPA	ALLM	31.589	0.079
	LUX-like	30.121	0.074
GRAPE (BH)	Hybrid	30.857	0.078
CEPGEN (LPAIR)	Suri-Yennie	32.464	0.083
	ALLM	27.570	0.058

safely used for predictions at the LHeC, if there are no appropriate ME Monte Carlo event generators available. The comparison of the differential cross-sections shown in Fig. 4 further confirms a very good description provided by the EPA approach. These results also confirm that the two-photon production of muon (and electron) pairs will serve as an excellent calibration candle to allow precision studies of the $\gamma\gamma$ interactions at the LHeC discussed below.

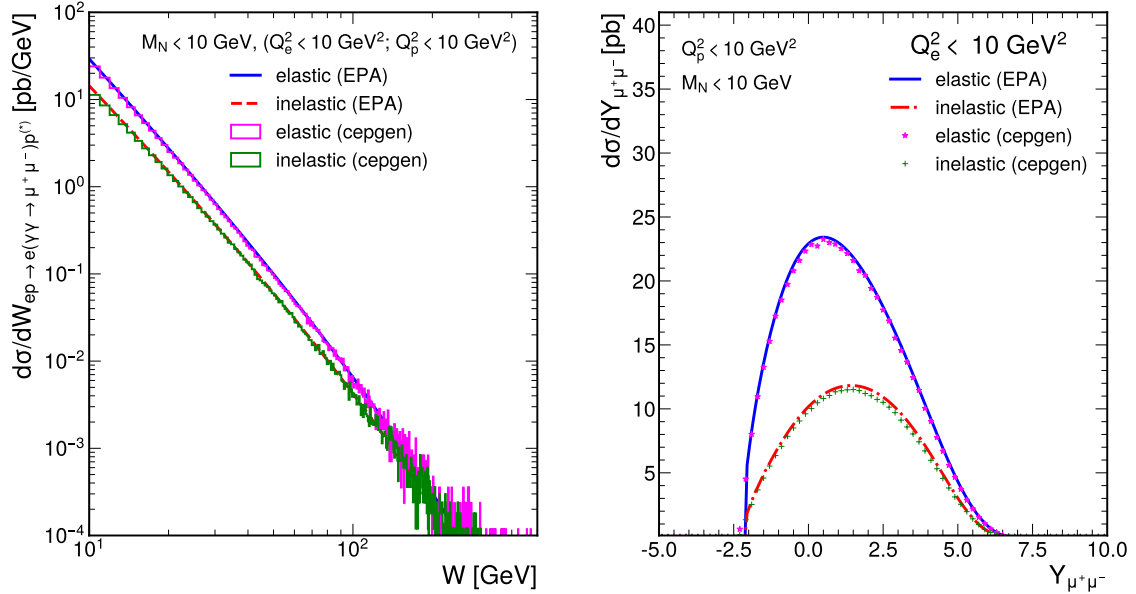


Figure 4. EPA results for the differential cross-sections in W and rapidity (where positive rapidity follows the direction of the proton), compared with those obtained from the CEPGEN implementation of LPAIR.

Finally, the differential cross sections in pseudo-rapidity, shown in Fig. 5, demonstrate that a large fraction of muon pairs will be produced within the acceptance of the tracking detector at the LHeC: $-4.6 < \eta_{\mu^-} < 5.2$ [17]. In addition, a significant fraction of these

events will be tagged by a detector of very forward protons, allowing a pure sample of elastic events to be obtained and a precise data-driven calibration of this proton detector.

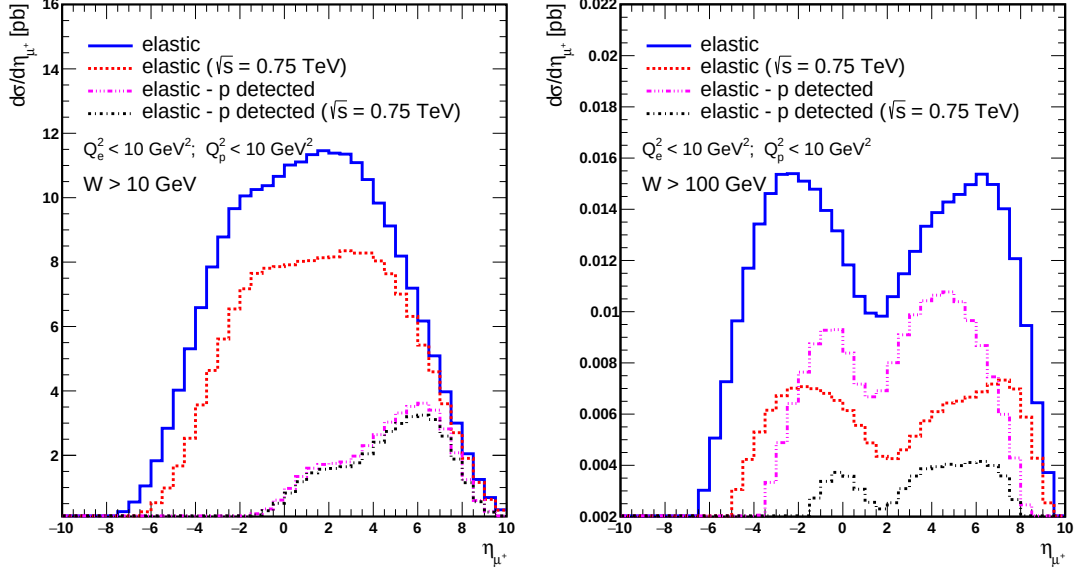


Figure 5. Elastic differential cross sections in pseudo-rapidity, calculated for $W > 10$ GeV (left) and for $W > 100$ GeV (right) using the CEPGEN implementation of LPAIR. An additional cut $-4.6 < \eta_{\mu^-} < 5.2$ is implemented for the tagged case assuming the proton detection corresponding to $0.01 < y_p < 0.1$.

4 Photon-photon factory of τ pairs

Table 3. Elastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 = 10$ and 100 GeV. Cross section values calculated using GRAPE and LPAIR (with dipole electromagnetic form factors) are presented for comparison.

σ^{el} (pb)	$Q_{e,p}^2 < 10 \text{ GeV}^2$	
	$W_0 = 10 \text{ GeV}$	$W_0 = 100 \text{ GeV}$
EPA	46.03	0.111
CepGen (LPAIR, dipole FF)	45.52	0.109
GRAPE (BH)	46.30	0.111
GRAPE (BH+ISR)	46.88	0.113
GRAPE (EW)	46.53	0.111
GRAPE (EW+ISR)	47.05	0.114

Due to the high statistics of τ pair production at LHeC, one can expect that it will probe the a_τ with high precision. A schematic diagram representing the two-photon production of τ pairs at the LHeC is shown in Fig. 8.

Table 4. Inelastic $\tau^+\tau^-$ production cross sections (in pb) at LHeC calculated using the EPA approach, for a minimum two-photon invariant mass of $W_0 = 10$ GeV. Cross section values calculated using GRAPE and the CepGen implementation of the LPAIR process (with Suri-Yennie proton structure functions) are presented for comparison

σ^{inel} (pb)	Structure func. modeling	$Q_{e,p}^2 < 10 \text{ GeV}^2, M_N < 3 \text{ GeV}$	
		$W_0 = 10 \text{ GeV}$	$W_0 = 100 \text{ GeV}$
EPA	ALLM	12.20	0.046
	LUX-like	11.63	0.043
CepGen (LPAIR)	Suri-Yennie	12.74	0.048
GRAPE (BH)	Hybrid	11.52	0.045
GRAPE (BH + ISR)	Hybrid	12.53	0.047
GRAPE (EW)	Hybrid	11.54	0.045
GRAPE (EW + ISR)	Hybrid	12.77	0.047

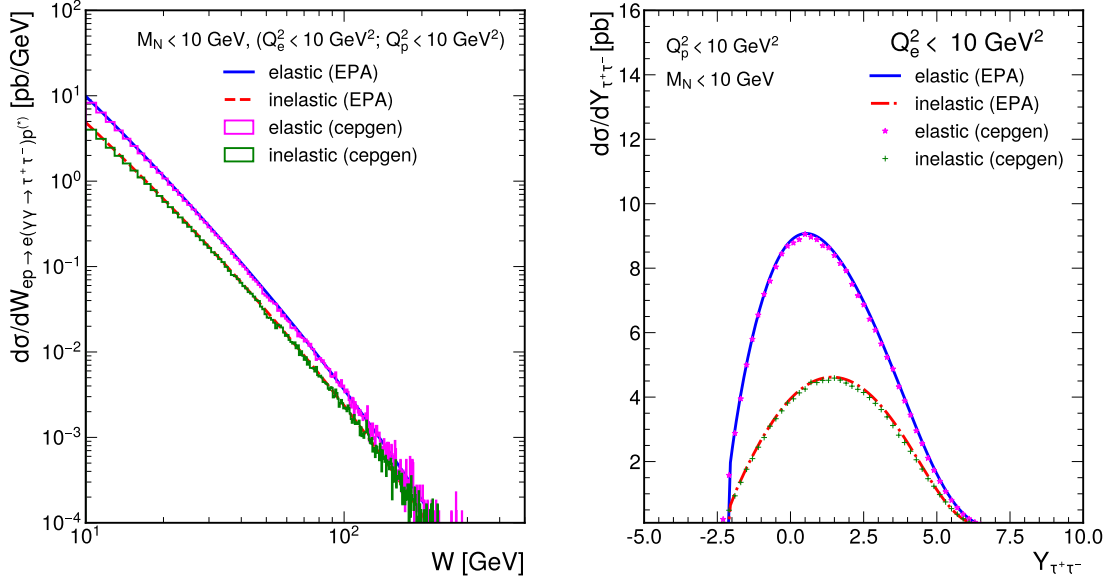


Figure 6. Central system invariant mass (left), and rapidity dependence for $ep \rightarrow ep^{(*)}(\gamma\gamma \rightarrow \tau^+\tau^-)$ process cross section at LHeC energies, for elastic (top), and inelastic (bottom) photon emissions from the proton side. Both the EPA (lines) approach and CepGen/LPAIR unweighted events (histograms) are shown for comparison. The distributions are shown by assuming $M_N < 10$ GeV for the mass of dissociated system and $Q_e^2 < 10 \text{ GeV}^2$ and $Q_p^2 < 10 \text{ GeV}^2$ for the electron/proton vertices.

In the specific decay channel, both leptonic decays of the τ particle (with a branching fraction of 35%) and hadronic decays of the τ particle (with an approximate branching fraction of 45%) could be considered (see Fig. 8). The τ lepton can undergo leptonic decay, where it decays into a charged lepton (ℓ), a corresponding neutrino (ν_ℓ), and a tau neutrino (ν_τ). It can also undergo hadronic decay, where it decays into a hadronic system consisting

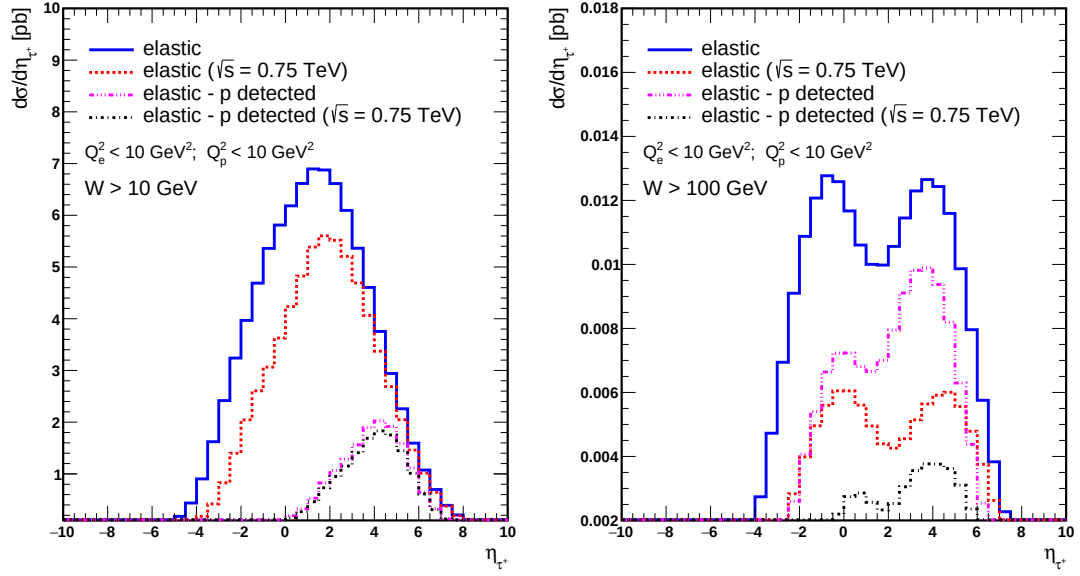


Figure 7. Elastic differential cross sections $d\sigma/d\eta_{\tau+}$ calculated for $W > 10$ (left) and > 100 GeV (right) using the CEPGEN implementation of the LPAIR process. The results are shown for the two center of mass energies of 1.2 and 0.75 TeV. For the ‘elastic - p detected’ case, an additional cut $-4.0 < \eta_{\tau-} < 4.0$ is also considered.

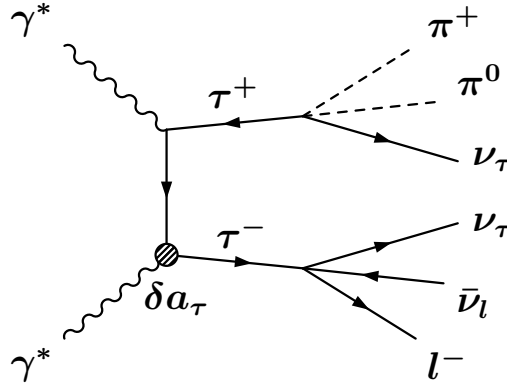


Figure 8. Schematic diagrams for the two-photon production of τ pairs ($\gamma\gamma \rightarrow \tau^+\tau^-$) at the LHeC. New Physics can modify tau-photon ($\tau-\gamma$) couplings affecting the magnetic moment. Both leptonic and hadronic τ decays are shown.

of charged pions ($\pi^\pm$), a neutral pion (π^0), and a tau neutrino (ν_τ).

In terms of background, the main sources of SM background includes dielectron and dimuon production through the process $\gamma\gamma \rightarrow \ell^+\ell^-$, exclusive photo production of W boson pairs through the process $\gamma\gamma \rightarrow W^+W^-$, and diquark production through the process

$\gamma\gamma \rightarrow q\bar{q}$. Considering the specific signature of the signal topology, it is possible to suppress the contributions of SM backgrounds. For example, in the signal scenario shown in Fig. 8, one can suppress the background from processes such as $\gamma\gamma \rightarrow \ell^+\ell^-$ and $\gamma\gamma \rightarrow W^+W^-$ by imposing a selection criterion that requires exactly one identified lepton in the event. This would effectively reduce the back-to-back lepton pairs originating from these background processes.

5 Exclusive production of W -boson pairs

The exclusive production of WW pairs via photon-photon interactions at LHeC is particularly well suited to study the beyond SM phenomena, and in particular, the anomalous triple γWW and quartic $\gamma\gamma WW$ gauge couplings. As can be seen from Feynman diagrams in Fig. 9, this process at leading order only involves diagrams with self-couplings of the electroweak gauge bosons, and hence, sensitive to anomalous gauge-boson interactions [18, 19].

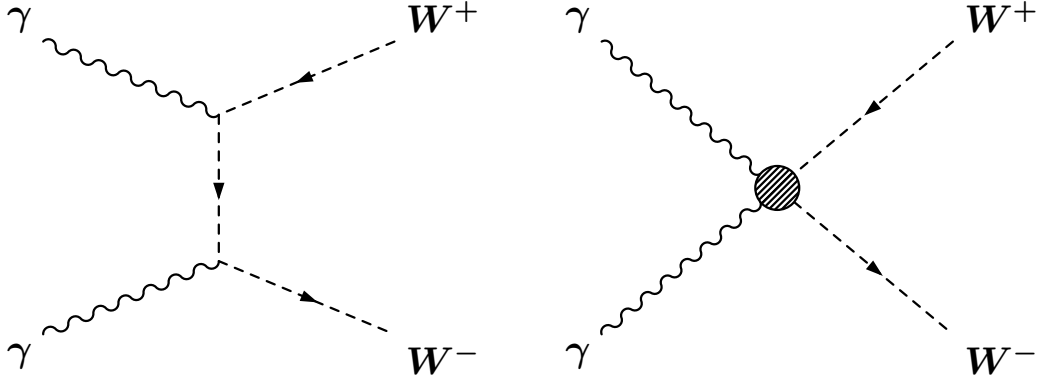


Figure 9. Leading-order Feynman diagrams contributing to the exclusive photon-induced production of W -boson pairs at the t -channel diagram (left) proceeding via the $\gamma\gamma W$ vertices, and a diagram with a quartic gauge coupling $\gamma\gamma WW$ (right).

Extensive research has been conducted on the production of W boson pairs through photon-photon interactions at the LHC as well. Very recent experimental efforts by ATLAS [20, 21] and CMS [22–24] Collaborations have been done for the observation of photon-induced production of W -boson pairs in proton-proton collisions. The cross-section for the $pp(\gamma\gamma) \rightarrow p^{(*)}W^+W^-p^{(*)}$ process in the decay channel $W^+W^- \rightarrow e^\pm\nu\mu^\mp\nu$ is measured to be ~ 3 fb, which is in agreement with the SM theoretical predictions [20].

The exclusive W boson pair production cross sections at the LHeC, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the elastic process as well as inelastic production with $M_N < 10, 50$ and 300 GeV is shown in Fig. 5.

As can be seen from this plot, at the LHeC and for very high photon virtuality, the cross-section for the two-photon exclusive WW pair production, even if it is small, still

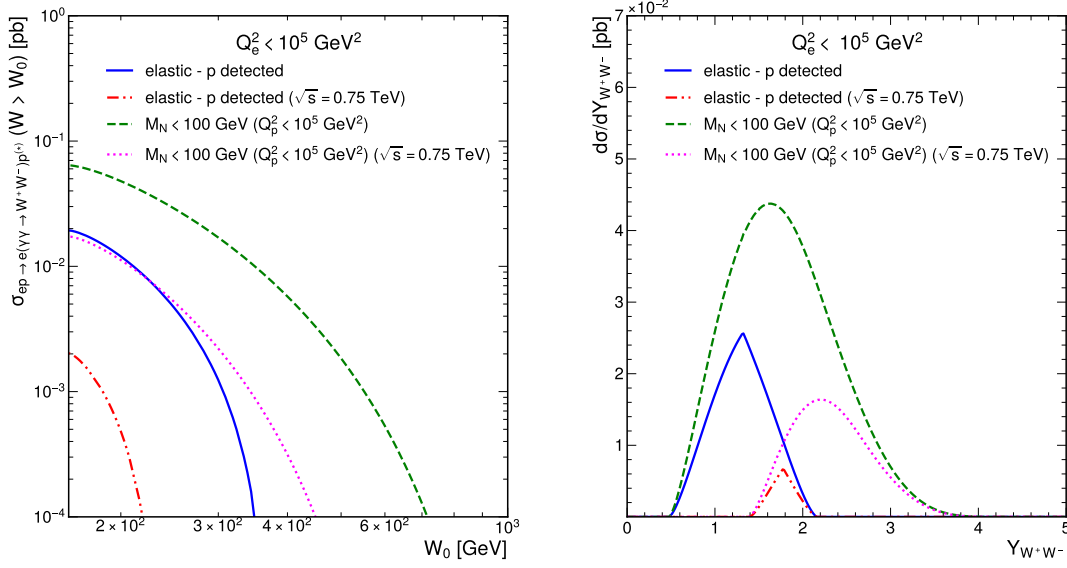


Figure 10. (left) Integrated W^+W^- boson pair production cross sections via photon-photon interaction at the LHeC, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{W^+W^-}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the tagged elastic process ($0.01 < y_p < 0.1$) as well as the inelastic production. Additionally, constraints on the diffractive system mass $M_N < 100$ GeV are applied for the inelastic case. The corresponding results for LHeC working at the center-of-mass energy of $\sqrt{s} = 0.75$ TeV are also shown. (right) Distribution for the differential cross sections $d\sigma/dY_{W^+W^-}$ for the W^+W^- boson pair production.

considerable, making LHeC a unique place to study the $\gamma\gamma \rightarrow W^+W^-$ process. This is mostly due to the favorable experimental conditions at LHeC, and the negligible event pileup as well.

Several signal typologies of $\gamma\gamma \rightarrow W^+W^-$ could be studied at the LHeC. Therefore, different selection criteria should be used to identify signal events and suppress relevant backgrounds. Among these typologies, the fully leptonic decays (involving muon and electron) of the W boson are particularly promising as they are considered potentially background-free. The requirement of two opposite-sign and same-flavor leptons ($ee/\mu\mu$) with sufficiently high transverse momentum can significantly suppress the backgrounds. Another interesting signal topology would involve the requirement of lepton pairs with different flavors, $W^+W^- \rightarrow e^\pm \nu \mu^\mp \nu$.

The dominant background at the LHeC in the context of WW pair production is typically reduced by selecting lepton pairs of different flavors, specifically $e\mu$ pairs, in order to suppress background processes such as $\gamma\gamma \rightarrow \ell\ell$ production. The small contribution from $\tau^+\tau^-$ processes with leptonic τ decays can still remain as one source of background. However, the contribution of this process decreases rapidly as the transverse momentum of the dilepton system increases [20]. By imposing higher requirements on the transverse momentum, this background can be further suppressed. In contrast to the LHC, the Drell-Yan (DY) production, which contributes to WW production via pp collisions at the LHC, is

completely absent at the LHeC. The background from the quark- and gluon-induced WW production ($qq(gg) \rightarrow WW$) also does not contribute.

Therefore, the distinctive fully leptonic signature of $\gamma\gamma \rightarrow W^+W^- \rightarrow \ell^+\ell^-\nu\nu$ and the experimental conditions at LHeC would offer a highly sensitive test for exploring the quartic gauge boson couplings within the Standard Model [18, 25]. Exclusive W boson pair production at LHeC also encompasses the exploration of anomalous triple gauge boson couplings γWW (as illustrated in Fig. 9), as well, however, the expected sensitivities are estimated to be less than the quartic couplings [19].

6 Unique Z boson laboratory

The exclusive production of ZZ pairs through two-photon interactions at the LHeC could provide an excellent platform for investigating the anomalous quartic gauge couplings (AQGC), specifically those involving $\gamma\gamma ZZ$. Such anomalous coupling appears in many extensions of the Standard Model, and in particular, could directly reveal the exchange of a new, heavy boson [5, 26].

The integrated Z -boson pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{ZZ}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the elastic process as well as inelastic production with $M_N < 10, 50$ and 300 GeV is shown in Fig. 11.

In the context of the LHeC and its investigation into high-energy $\gamma\gamma$ interactions, the Standard Model cross-section for the neutral gauge boson production process $ep \rightarrow e\gamma^*\gamma^*p \rightarrow eZZp$ is anticipated to be notably small (see Fig. 11). This is primarily due to the fact that the subprocess $\gamma^*\gamma^* \rightarrow ZZ$ is not allowed at the tree level but only occurs at the loop level, rendering it negligible. Consequently, the detection of even a few events in such a process could therefore reveal the existence of the AQGC and would serve as a significant hint of New Physics beyond the Standard Model. Without going into the details, we should stress that anomalous couplings can be introduced by building an effective Lagrangian that models low-energy behavior of a wide class of possible extensions of the SM. Considering a specific Lagrangian, the ZZ cross-sections can be written as a function of the related anomalous parameters. Then, one can indeed investigate the sensitivity to the AQGC using, for example, the di-lepton signature, $\gamma\gamma \rightarrow ZZ \rightarrow \ell^+\ell^-jj$. Finally, one can calculate the upper limits on the cross-section, which can be easily converted to limits on anomalous quartic couplings [5].

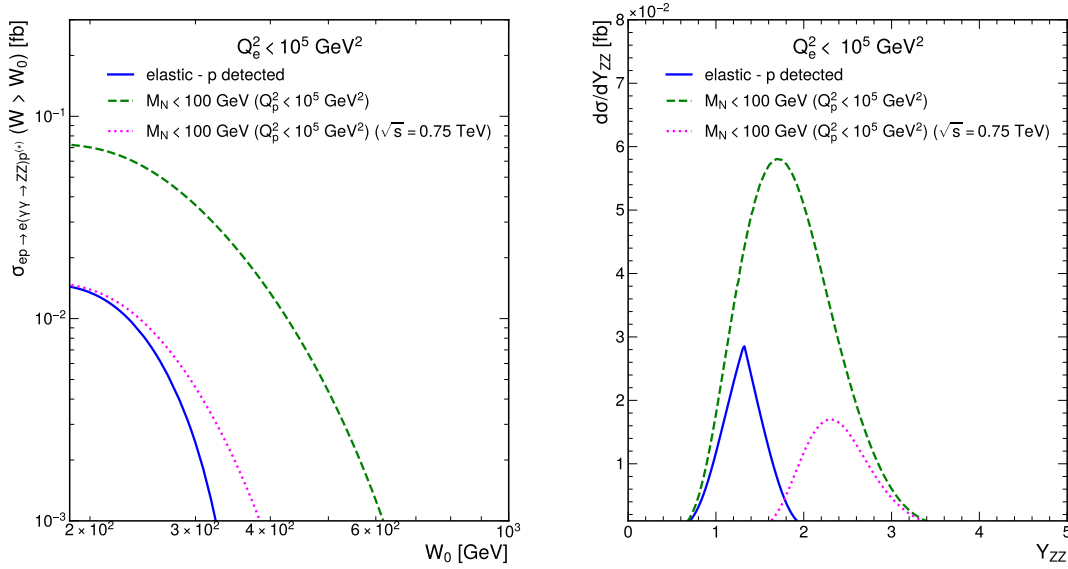


Figure 11. (left) Integrated Z-boson pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{ZZ}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , through the tagged elastic process ($0.01 < y_p < 0.1$) as well as inelastic production with $M_N < 100$ GeV. The corresponding inelastic result for LHeC working at the center-of-mass energy of $\sqrt{s} = 0.75$ TeV is also shown. (right) Differential cross section $d\sigma/dY_{ZZ}$ as a function of the rapidity $Y_{\tilde{\ell}^+ \tilde{\ell}^-}$.

7 Photoproduction of top quark-antiquark pairs

The calculated cross-sections for the exclusive production of top quark-antiquark pairs based on the EPA, is shown in Fig. 12 for both tagged elastic and inelastic processes.

The cross-section values calculated using the EPA approach for a minimum two-photon invariant mass of $W_0 \sim 2M_{\text{top}}$ are $\sigma^{\text{el}} = 1.889 \times 10^{-5}$ pb for tagged elastic production with $Q_e^2 < 10^5$ GeV 2 and $Q_p^2 < 10$ GeV 2 ; and $\sigma^{\text{inel}} = 3.048 \times 10^{-5}$ pb for the inelastic case, assuming a constraint on the diffractive system mass $M_N < 10$ GeV. Furthermore, when considering higher photon virtualities $Q_{e,p}^2 < 10^5$ GeV 2 and a diffractive system mass of $M_N < 300$ GeV, the calculated cross-section are estimated to be $\sigma^{\text{inel}} = 7.471 \times 10^{-5}$ pb for the inelastic case.

These results show that, despite the relatively small cross-section values, the clean experimental environment of the LHeC and the high photon-photon luminosity make the study of exclusive top quark pair production a valuable probe for both SM and beyond SM physics.

<https://ar5iv.labs.arxiv.org/html/2210.13884> <https://inspirehep.net/literature/2170538>
Search for central exclusive production of top quark pairs with the CMS and TOTEM experiments

<https://ar5iv.labs.arxiv.org/html/2007.04565> <https://inspirehep.net/literature/1806036>

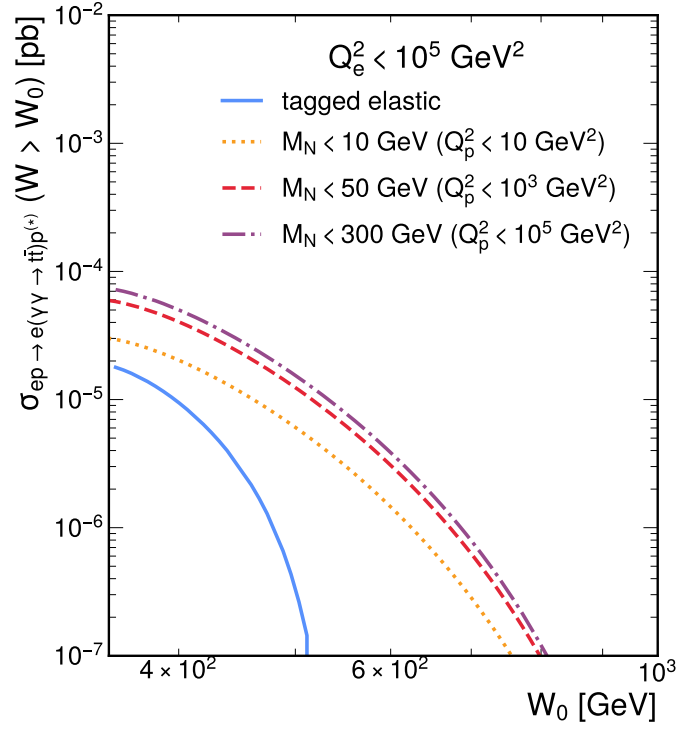


Figure 12. (Updated by exact EPA) Integrated exclusive top quark pair production cross sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{t\bar{t}}(W)$, as a function of the minimal $\gamma\gamma$ center-of-mass energy W_0 for both tagged elastic and inelastic production processes. Additional constraints on the diffractive system mass $M_N < 10, 50$, and 300 GeV are also considered for the inelastic case.

Top quark pair production in the exclusive processes at the LHC

8 Two-photon exclusive production of supersymmetric pairs

The investigation into the production of non-strongly interacting supersymmetric (SUSY) particles, specifically pairs of sleptons $\tilde{\ell}$, through photon-photon ($\gamma\gamma$) collisions at the LHeC hold considerable interest. The anticipated high luminosity at the LHeC, coupled with its clean experimental conditions, provides a unique opportunity to study these processes with improved statistical significance and reduced background noise.

The Feynman diagram representing the exclusive pair production of sleptons at LHeC, $ep \rightarrow e(\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-)p$, is shown in Fig. 13. The sleptons decay to their standard model partners ℓ and the lightest neutralino dark matter $\tilde{\chi}_1^0$, resulting in a very clean final state: two leptons (ee or $e\mu$ or $\mu\mu$) of opposite charge detected in the central region, and missing momentum due to escaping neutralinos. Additionally, the final state electron/proton, scattered at an almost zero-degree angle, is detectable at the dedicated very forward detectors (VFDs). Experimentally speaking, given that the beam energy is known, the measurements of the outgoing proton kinematics determine the colliding photon momenta and, consequently, the missing momentum. Very forward proton detectors at LHeC can be utilized to measure the photon energies, enabling a direct determination of the masses of the lightest SUSY particle, sleptons, with considerable resolution.

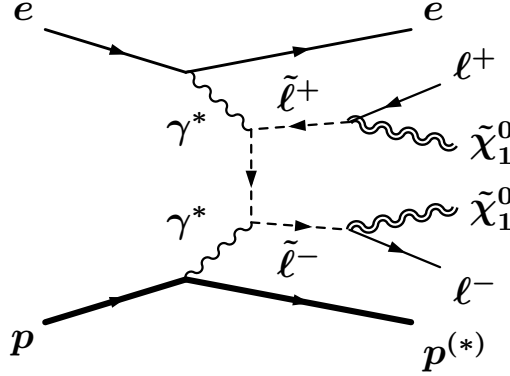


Figure 13. Exclusive pair production of scalar leptons (sleptons) $\tilde{\ell}$ at the LHeC, decaying to their standard model partners ℓ and the lightest neutralino dark matter $\tilde{\chi}_1^0$.

The integrated production cross sections of sleptons pairs, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tilde{\ell}^+\tilde{\ell}^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 , both for the elastic and inelastic processes are shown in Fig. ???. For the case of inelastic, we considered the $M_N < 10$ GeV. The integrated production cross sections are calculated for a specified value of $m_{\tilde{\ell}} = 100$ GeV. The cross-section does not exhibit logarithmic enhancement (see Eq. B.5), indicating that the production rate for sleptons is expected to be low for larger values of $m_{\tilde{\ell}}$. Experimental search for the supersymmetric particles could impose limits on their masses [27]. In Ref. [28], a novel search strategy is proposed, aiming to utilize the LHC as a photon

collider to improve sensitivity to the production of scalar leptons ($\tilde{\ell}$) in a compressed mass scenario, which involves masses ranging from 15 to 60 GeV above that of the neutralino dark matter ($\tilde{\chi}_1^0$). The primary outcomes of the study reveal increased sensitivity in comparison to LEP, expanding the detection capabilities to slepton masses up to 220 GeV for $15 \lesssim \Delta m(\tilde{\ell}, \tilde{\chi}_1^0) \lesssim 60$ GeV, considering 100 fb^{-1} of 13 TeV proton collisions.

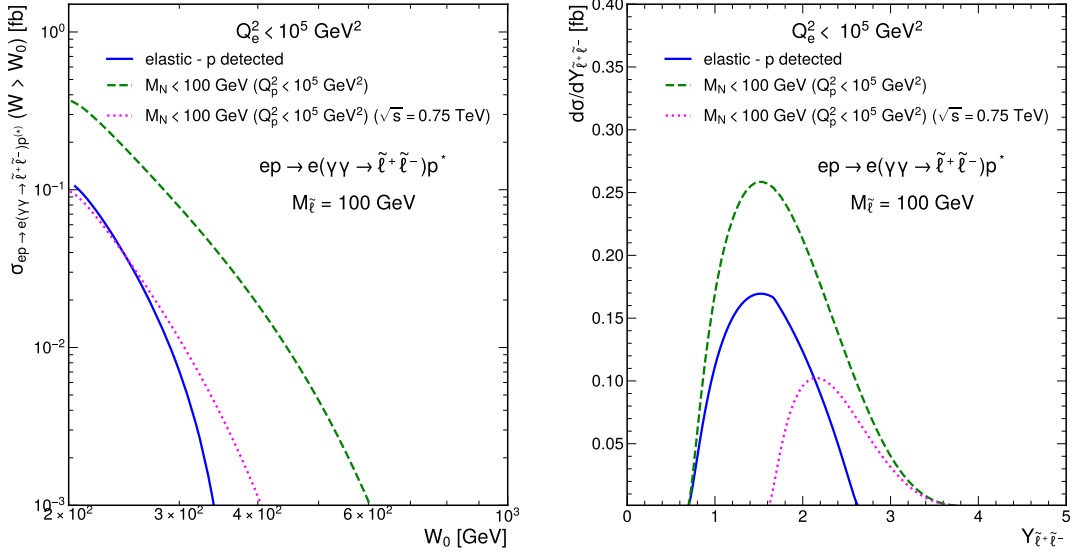


Figure 14. (left) Integrated slepton ($m_{\tilde{\ell}} = 100 \text{ GeV}$) pair production cross-sections, $\int_{W_0}^{\sqrt{s}} dW S_{\gamma\gamma} \sigma_{\tilde{\ell}^+\tilde{\ell}^-}$, as a function of the minimal $\gamma\gamma$ center-of-mass energy, W_0 . (right) Differential cross section $d\sigma/dY_{\tilde{\ell}^+\tilde{\ell}^-}$ as a function of the slepton rapidity $Y_{\tilde{\ell}^+\tilde{\ell}^-}$ for a given value of $M_{\tilde{\ell}} = 100 \text{ GeV}$. Distributions are shown for the tagged elastic process ($0.01 < y_p < 0.2$), and inelastic cases. For the latter, an upper limit $M_N < 100 \text{ GeV}$ is applied to the diffractive system mass.

Despite the anticipated high $\gamma\gamma$ luminosity at LHeC, the production cross-section of sleptons via photon-photon fusion is relatively low (see Fig. ??). This is partially compensated not only by the clean experimental conditions but also by the simple topology of the initial state and the potentially clean environment of the final state. Therefore, LHeC would be a perfect candidate for novel and complementary searches of sleptons using photon-photon collisions.

Considering the signal topology as $\gamma\gamma \rightarrow \tilde{\ell}^+\tilde{\ell}^-$ in which sleptons decay to their standard model partners ℓ and the lightest neutralino dark matter $\tilde{\chi}_1^0$, the expected dominant background primarily arises from the exclusive two-photon production of pairs of W bosons, followed by leptonic decays mode ($\gamma\gamma \rightarrow W^+W^-$), and the lepton pairs production ($\gamma\gamma \rightarrow \ell^+\ell^-$) as well.

Several considerations regarding the discrimination of signals from backgrounds merit discussion. Given the substantial increase in the cross sections for forward-backward scat-

tering, it is possible to implement cuts on the rapidity $|y|$ and the transverse momentum p_T of the final state particles. Additionally, the invariant mass of the produced slepton pairs should satisfy $M > 2m$ [29]. Unlike in signal events where the transverse momenta of the final state W boson or ℓ differ due to the escaping invisible neutralino, in $\gamma\gamma \rightarrow \ell^+\ell^-$ and $\gamma\gamma \rightarrow W^+W^-$ events, the transverse momenta of the W bosons or ℓ are balanced. Suggestively, by selectively rejecting events where the transverse momenta of the W or ℓ particles balance within a few GeV, the backgrounds are substantially reduced while the signals experience minimal impact. As discussed earlier, with an increase in the slepton mass, $m(\tilde{\ell})$, the $\gamma\gamma \rightarrow \tilde{\ell}\tilde{\ell}$ cross-section decreases, practically leading to a statistical limitation in the search at LHeC. However, signals with larger $m(\tilde{\ell})$ could become more distinguishable from the $\gamma\gamma \rightarrow W^+W^-$ background, benefiting from a clearer separation from the W boson mass [28]. The $\gamma\gamma \rightarrow \ell^+\ell^-$ background could be suppressed by requiring large missing energy as well.

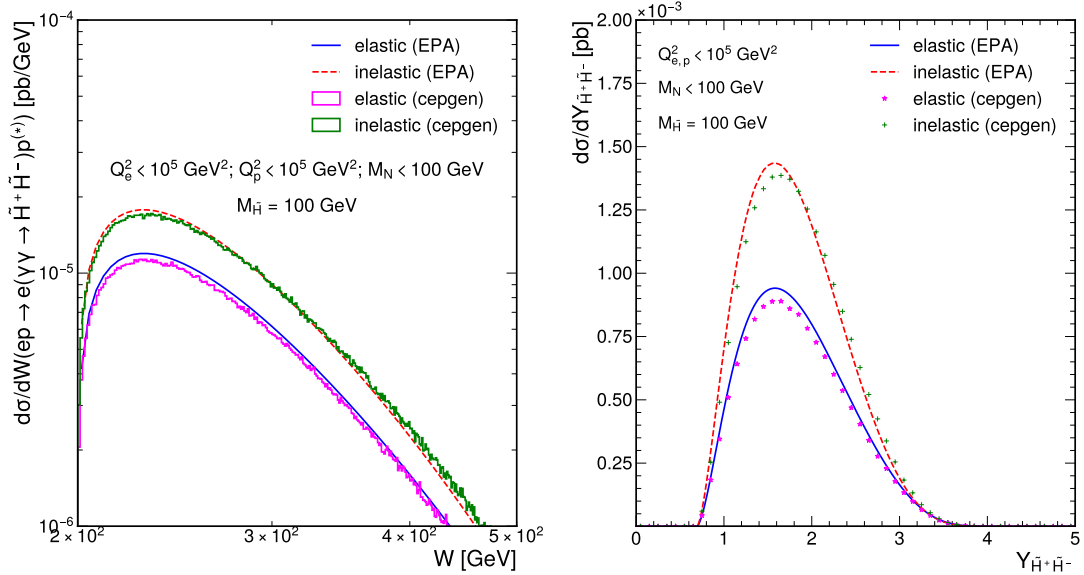


Figure 15. Differential cross sections for higgsino pair production: (left) $d\sigma/dW$ as a function of the photon-photon center-of-mass energy W ; (right) $d\sigma/dY_{\tilde{H}^+\tilde{H}^-}$ as a function of the higgsinos rapidity $Y_{\tilde{H}^+\tilde{H}^-}$. For the inelastic case, a photon virtualities upper limits $Q_{e,p}^2 < 10^5 \text{ GeV}^2$ is applied. An upper limit $M_N < 100 \text{ GeV}$ is applied to the diffractive system mass.

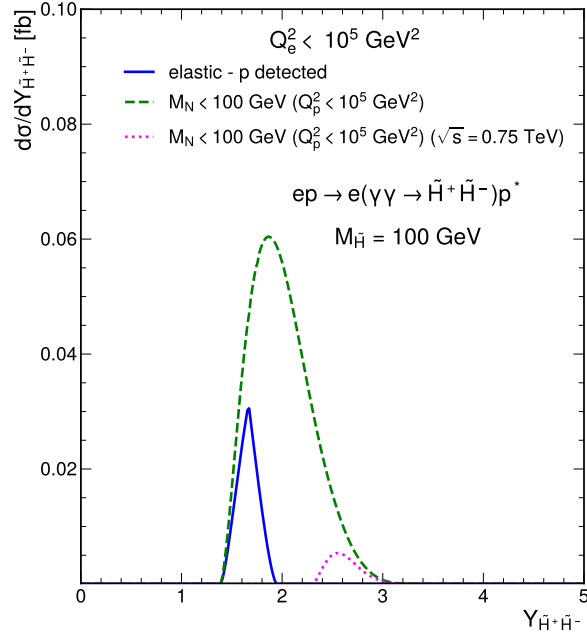


Figure 16. Differential cross section $d\sigma/dY_{\tilde{H}^+\tilde{H}^-}$ as a function of the higgsino rapidity $Y_{\tilde{H}^+\tilde{H}^-}$ for a given value of $M_{\tilde{H}} = 100$ GeV. Distribution is shown for the tagged elastic process ($0.01 < y_p < 0.2$), and inelastic cases. For the latter, an upper limit $M_N < 100$ GeV is applied to the diffractive system mass.

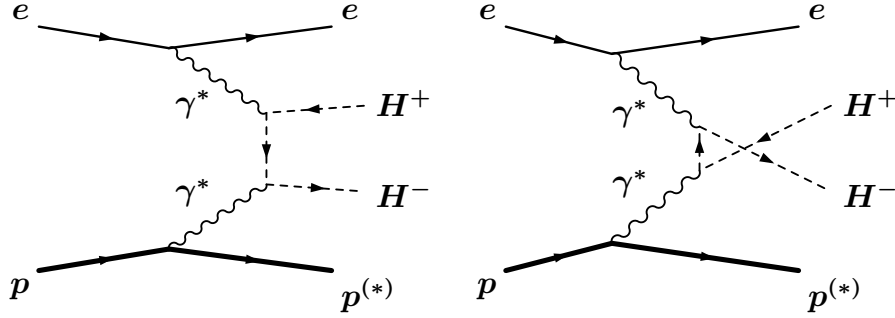


Figure 17. Feynman diagrams for the t and u -channel two-photon production of higgsinos at the LHeC.

9 Light-by-light scattering

Cross section of $\gamma\gamma \rightarrow \gamma\gamma$ needs to be calculated to have the statistics.

Waiting for Laurent to provide $\sigma_{\gamma\gamma \rightarrow \gamma\gamma}$.

- Short introduction and the main goal + references

Table 5. Cross-sections at the generator level for a minimum two-photon invariant mass of $W_0 = 200$ and 400 GeV for the $\gamma\gamma \rightarrow \tilde{H}^+ \tilde{H}^-$ process using EPA approach, GRAPE and CEPGEN (with the LPAIR process) event generators. Values are presented for two higgsino mass scenarios, $M_{\tilde{H}} = 100$ and 200 GeV, assuming a constraint on the diffractive system mass $M_N < 100$ GeV. **Numbers are updated for $M_N < 100$ GeV.**

Generator	$Q_{e,p}^2 < 10^5 \text{ GeV}^2$			
	$W_0 = 200 \text{ GeV}$		$W_0 = 400 \text{ GeV}$	
	σ^{el} (pb)	σ^{inel} (pb)	σ^{el} (pb)	σ^{inel} (pb)
EPA	1.322×10^{-3}	1.938×10^{-3}	3.383×10^{-5}	4.484×10^{-5}
LPAIR (CEPGEN [SURI-YENNIE])	1.270×10^{-3}	2.042×10^{-3}	3.258×10^{-5}	5.215×10^{-5}
GRAPE (BH)	1.298×10^{-3}	1.698×10^{-3}	3.343×10^{-5}	4.448×10^{-5}
GRAPE (EW)	1.375×10^{-3}	1.741×10^{-3}	3.756×10^{-5}	4.700×10^{-5}
GRAPE (BH+ISR)	1.335×10^{-3}	2.014×10^{-3}	3.469×10^{-5}	4.803×10^{-5}

- Feynman diagrams
- Backgrounds
- Short summary
 1. just what are the statistics of LbL at LHeC?
 2. (provided low backgrounds) how competitive is its BSM sensitivity for middle- and high-mass pairs wrt. the measurements elsewhere.

The interaction of two-photons is considered to be a purely quantum mechanics process that proceeds at leading-order (LO) in the fine structure constant and goes through the box diagram as shown in Fig. 18, where all charged particles including charged fermions (leptons and quarks) and boson ($W^\pm$) contribute to the amplitude.

Light-by-light (LbL) scattering has a tiny cross-section for optical photons [30, 31], but it is quite accessible for measurements at greater photon energies. Very recently this process was observed at the LHC in heavy-ion collisions by CMS [32, 33] and ATLAS [34] Collaborations at photon-photon center-of-mass energy of around $W_{\gamma\gamma} \approx 5$ GeV. Furthermore, LbL was sought for in a larger mass spectrum in proton-proton collisions [35]. Through the detection of forward protons kinematically consistent with the two-photon system energies at the order of $W_{\gamma\gamma} \approx 2$ TeV could probe anomalous production processes in this range. Several other studies for the cross-section measurements of this process at the HL-LHC and future collider FCC are available in literature [36–38].

Elastic LbL scattering, $\gamma\gamma \rightarrow \gamma\gamma$, can be measured in electromagnetic interactions of electron-proton at the LHeC as well, thanks to the large photon fluxes available at very high-energy collisions. **cross-section + statistics**

sensitivity to axion-like particles

<https://arxiv.org/abs/1904.10657>

[https://link.springer.com/article/10.1007/JHEP06\(2018\)131](https://link.springer.com/article/10.1007/JHEP06(2018)131)

[arXiv:1602.08088v1](https://arxiv.org/abs/1602.08088v1) [hep-ph] 25 Feb 2016

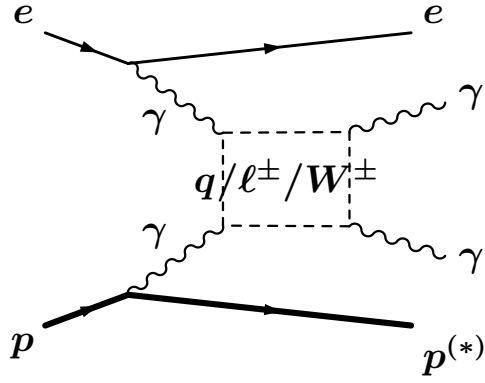


Figure 18. Feynman diagrams representing the elastic $\gamma\gamma \rightarrow \gamma\gamma$ collisions via the electron-proton interaction at the LHeC.

arXiv:1305.7142v3 [hep-ph] 25 Feb 2016
arXiv:1602.08088v1 [hep-ph] 25 Feb 2016

10 Production of the Higgs boson via photon-photon fusion

Future LHeC and its high-energy upgrade FCC-eh have emerged as potential candidates for a ‘Higgs factory’ experiment. These colliders offer the capability to produce Higgs bosons through the neutral- and charged-current weak boson fusion process (NC/CC WBF). The CC WBF $e + p \rightarrow \nu_e + H + j$ process is of great significance for the precise measurement of the Yukawa couplings [39–41]. The CC WBF process, which is considered to be the leading production mode of the Higgs boson at the LHeC, exhibits a significantly larger cross-section (~ 100 fb) compared to other production modes. The cross-section for Higgs boson production through the CC channel is approximately 5.3 times larger than the production cross-section corresponding to the NC channel at the LHeC with $E_e = 60$ GeV and $E_p = 7$ TeV. For the case of CC WBF, and for the decay channel $h \rightarrow b\bar{b}$, the cross-section $\sigma(ep \rightarrow H j \nu, H \rightarrow b\bar{b})$ is calculated to be at around 80 fb. The presence of a tagged forward jet has been identified as a critical factor in effectively reducing background interference in studies on the $Hb\bar{b}$ and $Hc\bar{c}$ couplings, and the azimuthal angle between the neutrino and the forward jet provides a sensitive probe of the anomalous HWW coupling at the LHeC.

The Feynman diagrams illustrating the Higgs photo-production process through W and charged fermion loops are depicted in Figure 19. In particular, it is worth noting that the dominant contributions to the fermion loop come from the top and bottom quarks.

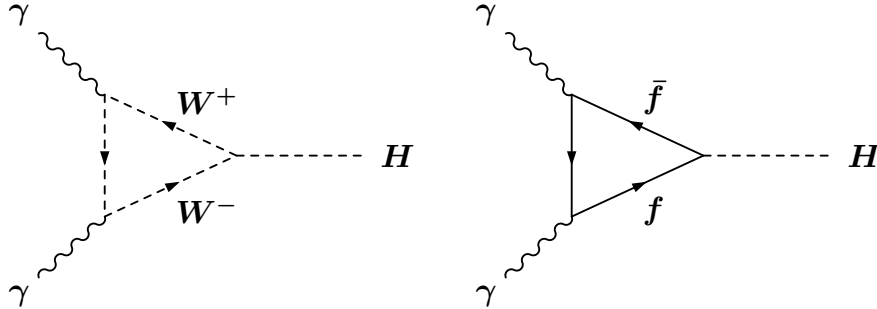


Figure 19. Feynman diagrams of the Higgs photo-production process through W (left) and charged fermion loops (right).

The photo-production cross-sections are estimated to be in the order $0.4 - 0.5$ fb for the electron beam energy of 60 GeV [42] at LHeC [needs to be verified as well], which are considerably smaller than the cross-sections for CC and NC WBF production. It is interesting that the Higgs production rate through photon-photon collision at the LHeC is still significant and comparable to that of FCC-ee/CEPC operating at the center of mass energy of 240 GeV and with an integrated luminosity of approximately 5 ab^{-1} . The estimated cross-section for Higgs two-photon production is 0.25 fb at FCC-ee/CEPC [43].

For the photon-photon production of the Higgs boson shown in Fig. 19, various decay

channels of the Higgs boson can be taken into account, including the main channels of $h \rightarrow b\bar{b}$, $h \rightarrow ZZ$ and $h \rightarrow W^+W^-$. For the $h \rightarrow b\bar{b}$ decay channel, the main background processes arise from $\gamma g \rightarrow b\bar{b}$ and $\gamma\gamma \rightarrow b\bar{b}$. As a result of the large difference between gluon and photon PDFs, the cross-sections for these background processes are orders of magnitude greater than that of the photo-production of the Higgs boson. Therefore, the utilization of the $h \rightarrow b\bar{b}$ decay channel for measuring the photon-photon production of the Higgs boson poses significant challenges.

In the case of the $h \rightarrow ZZ$ channel, even when considering the decay of ZZ into same-flavor and opposite-sign (SFOS) lepton pairs, which could help in suppressing the QCD background, there remains a significant difference between the signal and background cross-sections. This large discrepancy makes it nearly impossible to carry out precise measurements of Higgs photo-production in this particular channel.

In the $h \rightarrow W^+W^-$ channel, a possible approach is to consider the semi-leptonic decay of W bosons, where one W boson decays leptonically ($\ell\nu_\ell$) and the other decays hadronically into a final state with two jets (jj). The main background, in this case, arises from the process of $\gamma\gamma \rightarrow W^+W^-$ with semi-leptonic decays of the W bosons. The NC WBF process will also contribute as a background to the analysis of the Higgs boson production in the $h \rightarrow W^+W^-$ channel. It has been shown in the literature that even in the most favorable semi-leptonic W^+W^- channel which has an acceptable signal-to-background ratio, it is challenging to identify the photo-production of the Higgs boson due to an irreducible background from the photo-production of W^+W^- [42].

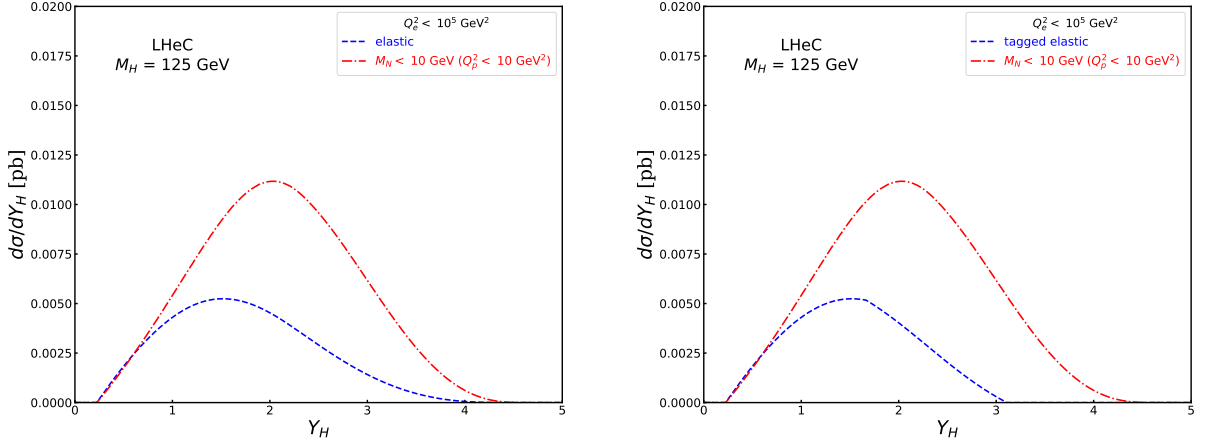


Figure 20. The differential cross section in rapidity, $d\sigma/dY$, for the production of Higgs boson ($M_H = 125$ GeV) and of $b\bar{b}$ pairs at $123 < W < 127$ GeV, for the tagged elastic case.

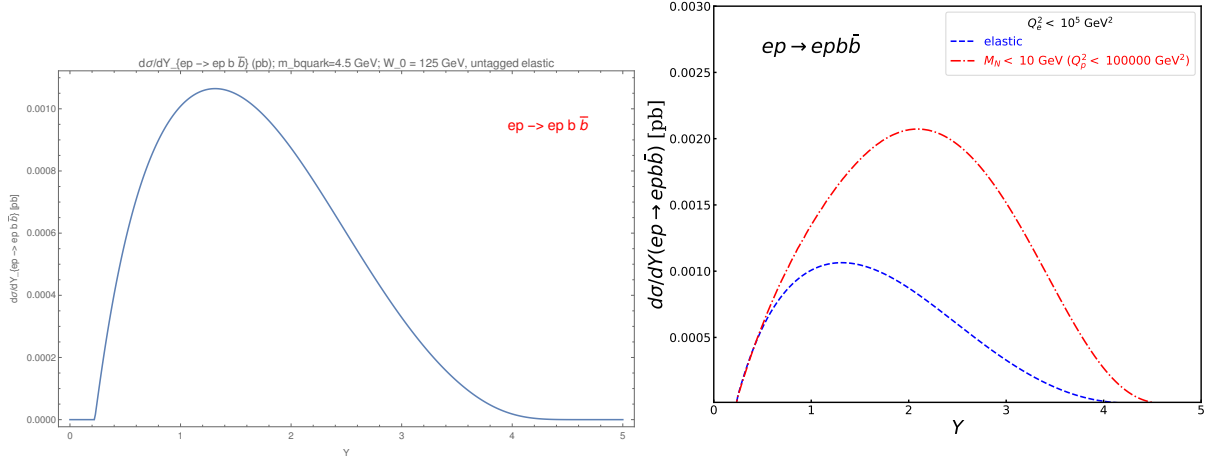


Figure 21. The differential cross section in rapidity, $d\sigma/dY$, for $ep \rightarrow epb\bar{b}$ for the inelastic and elastic case.

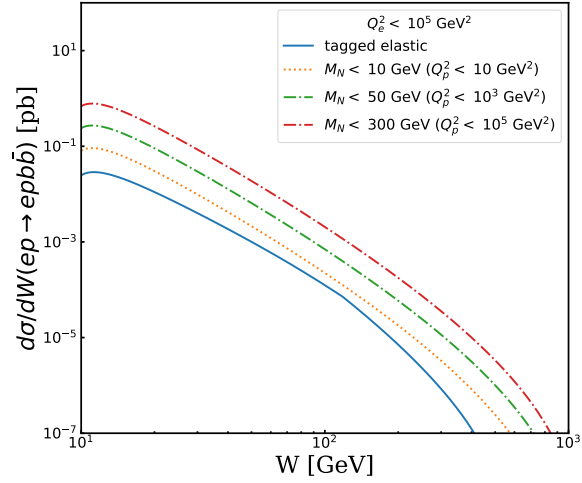


Figure 22. The differential cross section, $d\sigma/dW$, for $ep \rightarrow epb\bar{b}$ for the inelastic and elastic case.

11 Summary and conclusions

The first exploratory survey of the exclusive production of lepton pairs, pairs of photons as well as pairs of W and Z bosons via $\gamma\gamma$ processes at the LHeC reveals excellent prospects for studying such processes. The combination of very high luminosities available, advanced high-resolution detectors, and more importantly clean experimental environment at LHeC offers optimal conditions for further investigations into $\gamma\gamma$ processes.

A $\gamma\gamma$ luminosity spectrum

The luminosity spectrum $S_{\gamma\gamma}$ for $Q^2 \ll W^2$ is given by a convolution of electron and proton photon fluxes,

$$S_{\gamma\gamma} = \frac{2}{W} \int_{W^2/s}^1 dy_e \Phi_e(y_e) y_p \Phi_p(y_p), \quad (\text{A.1})$$

where $y_p = W^2/y_e s$, and therefore one can write

$$S_{\gamma\gamma} = \frac{2W}{s} \int_{W^2/s}^1 \frac{dy_e}{y_e} \Phi_e(y_e) \Phi_p\left(\frac{W^2}{y_e s}\right) \quad (\text{A.2})$$

and

$$d\sigma_{ep}/dW = S_{\gamma\gamma} \sigma_{\gamma\gamma} \quad (\text{A.3})$$

and

$$d\sigma_{ep}/dY = \frac{2}{s} \int_{W_0}^{\sqrt{s}} dW \sigma_{\gamma\gamma}(W) W \Phi_e(y_e) \Phi_p(y_p) = \frac{1}{s} \int_{s_0}^s dW^2 \sigma_{\gamma\gamma}(W) \Phi_e(y_e) \Phi_p(y_p), \quad (\text{A.4})$$

where $Y = \frac{1}{2} \ln [E_p y_p / (E_e y_e)]$ is the rapidity of the $\gamma\gamma$ system and $y_p = W \exp(Y)/(2E_p)$ and $y_e = W \exp(-Y)/(2E_e)$. During the integration one needs to make sure that the photon fluxes are equal to zero for y_p or $y_e > 1$.

For highly virtual photons one has to use the proper relation,

$$W^2 = -Q_e^2 - Q_p^2 + y_e y_p s + Q_e^2 (Q_p^2 + M_N^2 - M_p^2)/s, \quad (\text{A.5})$$

to calculate the photon-photon luminosity spectrum.

B $\gamma\gamma$ cross-sections

- The $\gamma\gamma$ cross-section for the production of W boson pairs near the threshold, below the $\gamma\gamma$ center-of-mass energy $W = 300$ GeV [44], is given by

$$\sigma(\gamma\gamma \rightarrow W^+ W^-)(W) \simeq \frac{19\pi(\hbar c)^2 \alpha^2}{2M_W^2} \sqrt{1 - \frac{4M_W^2}{W^2}}, \quad (\text{B.1})$$

where M_W is the mass of the W boson, and α is the fine-structure constant. Above $W = 300$ GeV, the cross-section is assumed to be constant and equal to

$$\sigma(\gamma\gamma \rightarrow W^+ W^-) \approx \frac{8\pi(\hbar c)^2 \alpha^2}{M_W^2}. \quad (\text{B.2})$$

- The $\gamma\gamma$ cross-section for the production of a charged fermion-antifermion pair is given by [45]:

$$\sigma(\gamma\gamma \rightarrow f^+ f^-)(W) = \frac{4\pi(\hbar c)^2 \alpha^2}{W^2} \beta \left(\frac{3 - \beta^4}{2\beta} \ln \left(\frac{1 + \beta}{1 - \beta} \right) - 2 + \beta^2 \right), \quad (\text{B.3})$$

where $\beta = \sqrt{1 - \frac{4m^2}{W^2}}$, and m is the fermion mass.

- The Higgs boson production cross-section in photon-photon collisions, $\sigma(\gamma\gamma \rightarrow H)$, can be calculated using the narrow resonance approximation, which is given by [43, 46, 47]:

$$\sigma(\gamma\gamma \rightarrow H)(W) \simeq 8\pi^2 \frac{\Gamma_{\gamma\gamma}(\hbar c)^2}{M_H} \frac{1}{\pi} \frac{M_H \Gamma}{(M_H^2 - W^2)^2 + M_H^2 \Gamma^2} \simeq 4\pi^2 \frac{\Gamma_{\gamma\gamma}(\hbar c)^2}{M_H^2} \delta(W - M_H), \quad (\text{B.4})$$

where M_H is the Higgs boson mass, $\Gamma_{\gamma\gamma}$ is the two-photon decay width, and Γ is the total decay width of the Higgs boson.

- The cross-section for slepton pair production via two-photon fusion ($\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-$) is given by [29, 45]:

$$\sigma(\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-)(W) = \frac{2\pi(\hbar c)^2 \alpha^2}{W^2} \beta \left[2 - \beta^2 - \frac{1 - \beta^4}{2\beta} \ln \left(\frac{1 + \beta}{1 - \beta} \right) \right], \quad (\text{B.5})$$

where $\beta = \sqrt{1 - \frac{4m_{\tilde{\ell}}^2}{W^2}}$, with $m_{\tilde{\ell}}$ being the mass of the slepton. Similarly, the production of charged Higgs boson pairs in photon-photon collisions ($\gamma\gamma \rightarrow H^+ H^-$) occurs at the same rate as slepton pair production, assuming equal masses for the sleptons and charged Higgs bosons.

Acknowledgments

L. Forthomme and K. Piotrkowski appreciate the financial support from the Polish National Agency for Academic Exchange (NAWA) under grant number BPN/PPO/2021/1/00011, and L. Forthomme is also supported by the NCN grant 2022/01/1/ST2/00022. H. Khanpour appreciates the financial support from NAWA under grant number BPN/ULM/2023/1/00160 and from the IDUB programme at the AGH University.

References

- [1] LHeC, FCC-HE STUDY GROUP collaboration, *The large hadron-electron collider at the HL-LHC*, *J. Phys. G* **48** (2021) 110501 [2007.14491].
- [2] F. Ahmadova et al., *The Large Hadron electron Collider as a bridge project for CERN*, 2503.17727.

- [3] K.D.J. André, B. Holzer, L. Forthomme and K. Piotrkowski, *An electron-hadron collider at the high-luminosity LHC*, [2503.20475](#).
- [4] K. Piotrkowski, *Tagging two photon production at the CERN LHC*, *Phys. Rev. D* **63** (2001) 071502 [[hep-ex/0009065](#)].
- [5] J. de Favereau de Jeneret, V. Lemaitre, Y. Liu, S. Oryn, T. Pierzchala, K. Piotrkowski et al., *High energy photon interactions at the LHC*, [0908.2020](#).
- [6] CMS collaboration, *Exclusive photon-photon production of muon pairs in proton-proton collisions at $\sqrt{s} = 7$ TeV*, *JHEP* **01** (2012) 052 [[1111.5536](#)].
- [7] CMS, TOTEM collaboration, *Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer*, *JHEP* **07** (2018) 153 [[1803.04496](#)].
- [8] ATLAS collaboration, *Observation and measurement of forward proton scattering in association with lepton pairs produced via the photon fusion mechanism at ATLAS*, *Phys. Rev. Lett.* **125** (2020) 261801 [[2009.14537](#)].
- [9] V.M. Budnev, I.F. Ginzburg, G.V. Meledin and V.G. Serbo, *The two photon particle production mechanism. physical problems. applications. equivalent photon approximation*, *Phys. Rept.* **15** (1975) 181.
- [10] H. Abramowicz, E.M. Levin, A. Levy and U. Maor, *A parametrization of $\sigma_T(\gamma^*p)$ above the resonance region $Q^2 \geq 0$* , *Phys. Lett. B* **269** (1991) 465.
- [11] H. Abramowicz and A. Levy, *The ALLM parameterization of $\sigma_{\text{tot}}(\gamma^*p)$: An update*, [hep-ph/9712415](#).
- [12] T.-J. Hou et al., *New CTEQ global analysis of quantum chromodynamics with high-precision data from the LHC*, *Phys. Rev. D* **103** (2021) 014013 [[1912.10053](#)].
- [13] NNPDF collaboration, *The path to proton structure at 1% accuracy*, *Eur. Phys. J. C* **82** (2022) 428 [[2109.02653](#)].
- [14] T. Abe, *GRAPE dilepton (version1.1): A generator for dilepton production in ep collisions*, *Comput. Phys. Commun.* **136** (2001) 126 [[hep-ph/0012029](#)].
- [15] S.P. Baranov, O. Duenger, H. Shooshtari and J.A.M. Vermaseren, *LPAIR: A generator for lepton pair production*, in *Workshop on Physics at HERA*, 1991.
- [16] L. Forthomme, *CepGen – a generic central exclusive processes event generator for hadron-hadron collisions*, *Comput. Phys. Commun.* **271** (2022) 108225 [[1808.06059](#)].
- [17] K.D.J. André et al., *An experiment for electron-hadron scattering at the LHC*, *Eur. Phys. J. C* **82** (2022) 40 [[2201.02436](#)].
- [18] T. Pierzchala and K. Piotrkowski, *Sensitivity to anomalous quartic gauge couplings in photon-photon interactions at the LHC*, *Nucl. Phys. B Proc. Suppl.* **179-180** (2008) 257 [[0807.1121](#)].
- [19] O. Kepka and C. Royon, *Anomalous $WW\gamma$ coupling in photon-induced processes using forward detectors at the LHC*, *Phys. Rev. D* **78** (2008) 073005 [[0808.0322](#)].
- [20] ATLAS collaboration, *Observation of photon-induced W^+W^- production in pp collisions at $\sqrt{s} = 13$ TeV using the ATLAS detector*, *Phys. Lett. B* **816** (2021) 136190 [[2010.04019](#)].

- [21] ATLAS collaboration, *Measurement of exclusive $\gamma\gamma \rightarrow W^+W^-$ production and search for exclusive Higgs boson production in pp collisions at $\sqrt{s} = 8$ TeV using the ATLAS detector*, *Phys. Rev. D* **94** (2016) 032011 [[1607.03745](#)].
- [22] CMS, TOTEM collaboration, *Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *JHEP* **07** (2023) 229 [[2211.16320](#)].
- [23] CMS collaboration, *Study of exclusive two-photon production of W^+W^- in pp collisions at $\sqrt{s} = 7$ TeV and constraints on anomalous quartic gauge couplings*, *JHEP* **07** (2013) 116 [[1305.5596](#)].
- [24] CMS collaboration, *Evidence for exclusive $\gamma\gamma \rightarrow W^+W^-$ production and constraints on anomalous quartic gauge couplings in pp collisions at $\sqrt{s} = 7$ and 8 TeV*, *JHEP* **08** (2016) 119 [[1604.04464](#)].
- [25] E. Chapon, C. Royon and O. Kepka, *Anomalous quartic $WW\gamma\gamma$, $ZZ\gamma\gamma$, and trilinear $WW\gamma$ couplings in two-photon processes at high luminosity at the LHC*, *Phys. Rev. D* **81** (2010) 074003 [[0912.5161](#)].
- [26] G. Belanger, F. Boudjema, Y. Kurihara, D. Perret-Gallix and A. Semenov, *Bosonic quartic couplings at LEP-2*, *Eur. Phys. J. C* **13** (2000) 283 [[hep-ph/9908254](#)].
- [27] G. Baur, K. Hencken, D. Trautmann, S. Sadovsky and Y. Kharlov, *Coherent $\gamma\gamma$ and γA interactions in very peripheral collisions at relativistic ion colliders*, *Phys. Rept.* **364** (2002) 359 [[hep-ph/0112211](#)].
- [28] L. Beresford and J. Liu, *Search strategy for sleptons and dark matter using the LHC as a photon collider*, *Phys. Rev. Lett.* **123** (2019) 141801 [[1811.06465](#)].
- [29] J. Ohnemus, T.F. Walsh and P.M. Zerwas, *$\gamma\gamma$ production of nonstrongly interacting SUSY particles at hadron colliders*, *Phys. Lett. B* **328** (1994) 369 [[hep-ph/9402302](#)].
- [30] D. d’Enterria and G.G. da Silveira, *Observing light-by-light scattering at the Large Hadron Collider*, *Phys. Rev. Lett.* **111** (2013) 080405 [[1305.7142](#)].
- [31] G.K. Krintiras, I. Grabowska-Bold, M. Khusek-Gawenda, E. Chapon, R. Chudasama and R. Granier de Cassagnac, *Light-by-light scattering cross-section measurements at LHC*, [2204.02845](#).
- [32] ATLAS collaboration, *Observation of light-by-light scattering in ultraperipheral Pb+Pb collisions with the ATLAS detector*, *Phys. Rev. Lett.* **123** (2019) 052001 [[1904.03536](#)].
- [33] ATLAS collaboration, *Evidence for light-by-light scattering in heavy-ion collisions with the ATLAS detector at the LHC*, *Nature Phys.* **13** (2017) 852 [[1702.01625](#)].
- [34] CMS collaboration, *Evidence for light-by-light scattering and searches for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV*, *Phys. Lett. B* **797** (2019) 134826 [[1810.04602](#)].
- [35] TOTEM, CMS collaboration, *First search for exclusive diphoton production at high mass with tagged protons in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. Lett.* **129** (2022) 011801 [[2110.05916](#)].
- [36] K.I. Beloborodov, T.A. Kharlamova and V.I. Telnov, *Problems of studying $\gamma\gamma \rightarrow \gamma\gamma$ scattering at e^+e^- and $\gamma\gamma$ colliders*, [2301.07089](#).
- [37] R.O. Coelho, V.P. Gonçalves, D.E. Martins and M. Rangel, *Exclusive and diffractive $\gamma\gamma$*

- production in PbPb collisions at the LHC, HL-LHC and FCC, *Eur. Phys. J. C* **80** (2020) 488 [[2002.03902](#)].
- [38] D. d’Enterria and G.G. da Silveira, *Measuring light-by-light scattering at the LHC and FCC*, in *16th conference on Elastic and Diffractive Scattering (EDS 15)*, 2, 2016 [[1602.08088](#)].
- [39] S. Behera, B. Brickwedde and M. Schott, *Studies on the $H \rightarrow b\bar{b}$ cross section measurement at the LHeC with a full detector simulation*, *Eur. Phys. J. C* **82** (2022) 878 [[2201.04037](#)].
- [40] G. Haghighat, R. Jafari, H. Khanpour and M. Mohammadi Najafabadi, *Exploring nonstandard $Hb\bar{b}$ interactions at future electron-proton colliders*, *Phys. Rev. D* **109** (2024) 095023 [[2305.05462](#)].
- [41] T. Han and B. Mellado, *Higgs boson searches and the $Hb\bar{b}$ coupling at the LHeC*, *Phys. Rev. D* **82** (2010) 016009 [[0909.2460](#)].
- [42] R. Li, X. Lv, B.-W. Wang, K. Wang and T. Xu, *Photo-production of the Higgs boson at the LHeC*, *Nucl. Phys. B* **958** (2020) 115134 [[1906.08969](#)].
- [43] I.R. Boyko, V.V. Bytev and A.S. Zhemchugov, *Two-photon physics at future electron-positron colliders*, *Chin. Phys. C* **47** (2023) 013001 [[2110.01426](#)].
- [44] M. Baillargeon, G. Belanger and F. Boudjema, *Electroweak physics issues at a high-energy photon collider*, in *Two-Photon Physics from DAPHNE to LEP200 and Beyond*, 5, 1994 [[hep-ph/9405359](#)].
- [45] S.J. Brodsky, T. Kinoshita and H. Terazawa, *Two photon mechanism of particle production by high-energy colliding beams*, *Phys. Rev. D* **4** (1971) 1532.
- [46] V.A. Khoze, A.D. Martin and M.G. Ryskin, *Prospects for new physics observations in diffractive processes at the LHC and Tevatron*, *Eur. Phys. J. C* **23** (2002) 311 [[hep-ph/0111078](#)].
- [47] D. d’Enterria and J.-P. Lansberg, *Study of Higgs boson production and its $b\bar{b}$ decay in $\gamma\gamma$ processes in proton-nucleus collisions at the LHC*, *Phys. Rev. D* **81** (2010) 014004 [[0909.3047](#)].